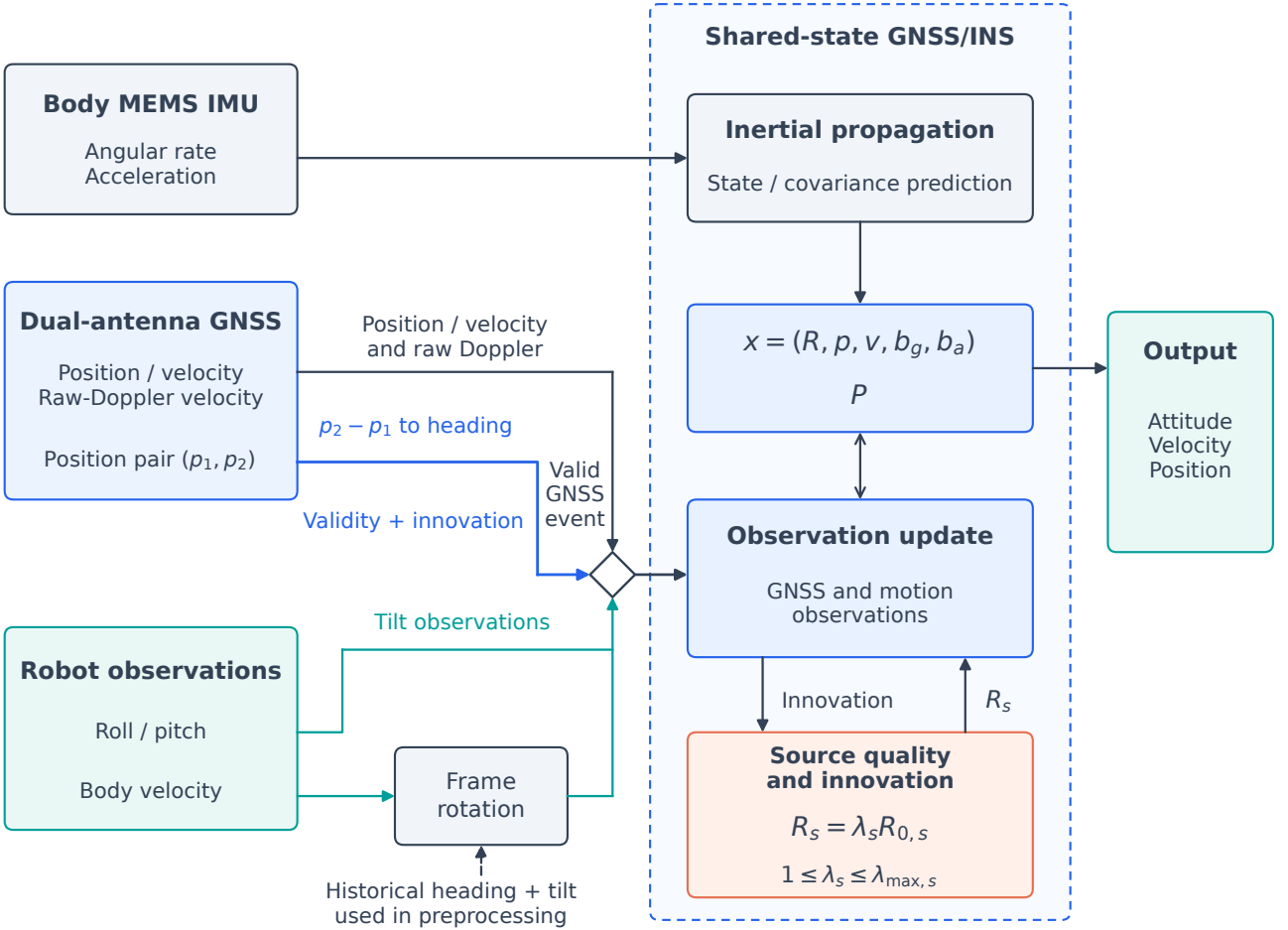
Shared-state GNSS/INS
Body MEMS IMU
Angular rate
Acceleration
Inertial propagation
State / covariance prediction
Dual-antenna GNSS
Position / velocity
Raw-Doppler velocity
Position pair (p_1, p_2)
Position / velocity
and raw Doppler
$p_2 - p_1$ to heading
Valid
GNSS
event
Validity + innovation
$x = (R, p, v, b_g, b_a)$
P
Output
Attitude
Velocity
Position
Observation update
GNSS and motion
observations
Tilt observations
Robot observations
Roll / pitch
Body velocity
Frame
rotation
Innovation
R_s
Source quality
and innovation
$R_s = \lambda_s R_{0,s}$
$1 \le \lambda_s \le \lambda_{\max,s}$
Historical heading + tilt
used in preprocessing